

2025 Mock $F = ma$ Exam

25 QUESTIONS - 75 MINUTES

INSTRUCTIONS

DO NOT OPEN THIS TEST UNTIL YOU ARE TOLD TO BEGIN

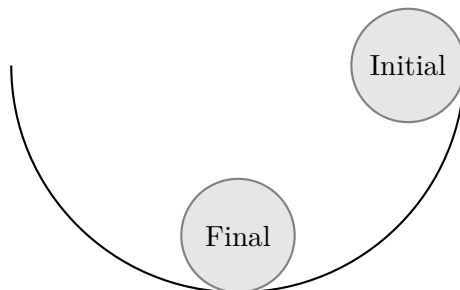
- Use $g = 10 \text{ N/kg}$ throughout this contest.
- You may write in this booklet of questions. However, you will not receive any credit for anything written in this booklet.
- Test under standard conditions, meaning that you must complete the test in 75 minutes in one sitting.
- This test contains 25 multiple-choice questions. **Your answer to each question must be marked on the [Google Forms](#) answer sheet that accompanies the test.**
- Correct answers will be awarded one point; incorrect answers and leaving an answer blank will be awarded zero points. There is no additional penalty for incorrect answers.
- A hand-held calculator may be used. Its memory must be cleared of data and programs. You may use only the basic functions found on a simple scientific calculator. Calculators may not be shared. Cell phones may not be used during the exam or while the exam papers are present. You may not use any tables, books, or collections of formulas. You must follow all of the contest rules on our [rules document](#).
- All questions are equally weighted but are not necessarily the same level of difficulty.
- This event is independent and not affiliated with, endorsed by, or sponsored by AAPT or the US Physics Team.
- **In order to maintain exam security, do not communicate any information about the questions (or their answers or solutions) on this contest until after 2:00 PM Pacific Standard Time on September 21, 2025.**

We acknowledge the following people for their contributions to this year's exam (in alphabetical order):

Leo Shao, Alan Wang

Special thanks to all test solvers who were able to give helpful feedback on the problems.

1. A uniform solid sphere of mass m and radius $r = \frac{R}{4}$ is released from rest at the edge of a fixed hemispherical bowl of radius R and rolls into the bowl without slipping. Find the sphere's angular displacement relative to its center when it reaches the bottom of the hemisphere.



- (A) $\frac{\pi}{2}$ (B) $\frac{3\pi}{4}$ (C) π (D) $\frac{5\pi}{4}$ (E) $\frac{3\pi}{2}$

$$v = \omega_{\text{CM}} r = \omega_{\text{bowl}} (R - r)$$

$$\omega_{\text{CM}} = \frac{\omega_{\text{bowl}} (R - r)}{r}$$

We can see that the angular velocity of the sphere with respect to its center is always $R/r - 1 = 3$ times the angular velocity of the sphere with respect to the bowl. Thus, after it rolls $\frac{\pi}{2}$ radians with respect to the bowl, it rolls $\frac{3\pi}{2}$ radians with respect to its center.

2. A ball of mass 5 kg is launched straight up from the ground with speed v_0 . The drag force on the ball is $-av_{\text{ball}}$ for $v_{\text{ball}} \leq v_1$ and $-bv_{\text{ball}}^2$ for $v_{\text{ball}} > v_1$, where $a = 5 \text{ N}\cdot\text{s}/\text{m}$, $b = 0.50 \text{ N}\cdot\text{s}^2/\text{m}^2$, $v_1 = 10 \text{ m/s}$, and v_{ball} is the instantaneous speed of the ball. Which of the following values of v_0 result in the highest speed at which the ball first hits the ground after being launched?

- (A) 9.8 m/s (B) 9.9 m/s (C) 10.0 m/s (D) 10.1 m/s (E) 10.2 m/s

The largest initial speed will give the largest maximum height above the ground, since any ball thrown at a larger speed will reach the speed of a ball thrown at a smaller speed after a positive nonzero displacement upwards. Dropping a ball from the largest height will provide the largest final speed if terminal velocity is not reached, which is the case here. Thus, the answer is E.

3. A system composed of many identical particles travels with a fixed total linear momentum $\vec{p}$. Which of the following conditions minimizes the total translational kinetic energy of the system?

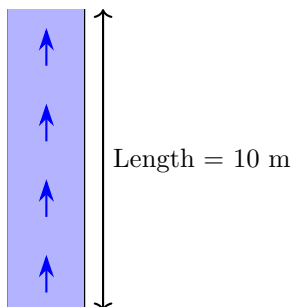
- (A) All of the particles travel with the same speed but not the same velocity.
 (B) All of the particles travel with the same velocity.
 (C) One of the particles travels with linear momentum $\vec{p}$, with all other particles at rest.
 (D) One particle remains at rest, with all other particles traveling at an identical velocity.
 (E) None of the above.

Consider the case where all particles are traveling at the same velocity. Now consider if one particle's velocity increases by Δv while another particle's velocity decreases by Δv . The change in kinetic energy is

$$\Delta KE = \frac{1}{2}m(v + \Delta v)^2 + \frac{1}{2}m(v - \Delta v)^2 - mv^2 = m(\Delta v)^2$$

which is always positive. Thus, the answer is B.

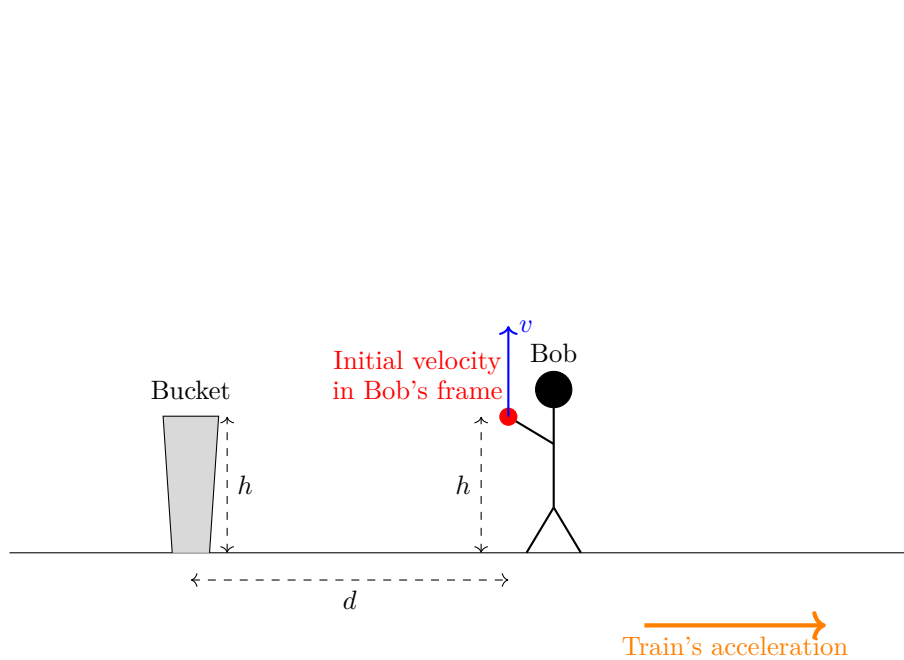
4. Water enters a vertical cylindrical pipe at speed 10 m/s. A motor then propels it upwards at a constant velocity through the pipe. Given that the pipe has length 10 m and that the motor generates power at a rate of 500 kW, find the radius of the pipe. The density of water is 1000 kg/m^3 . Assume that all of the power generated by the motor is transferred to the water.



- (A) 1.26 cm (B) 1.78 cm (C) 24.8 cm (D) 39.9 cm (E) 56.4 cm

Let ρ be the density of water. The motor provides gravitational potential energy to the water. The gravitational PE per unit time is $\rho g(\pi r^2 l)v$. This equals the power P . Solving for r , we obtain $r = \sqrt{\frac{P}{\rho g \pi l v}} = 39.9 \text{ cm}$.

5. Bob is standing still in a very tall train that is accelerating uniformly. A small bucket, also in the train, with a height h , is a distance d directly behind Bob. In his frame, Bob tosses a small ball at height h above the ground with velocity v straight up. Find the acceleration of the train necessary for the ball to land in the bucket. Neglect air resistance, assume that the bucket does not slip relative to the floor of the train or tip over, and that the ball does not hit the roof of the train.



- (A) $\frac{2dg^2}{v^2}$
 (B) $\frac{dg^2}{v^2}$
 (C) $\frac{dg^2}{2v^2}$
 (D) $\frac{dg^2}{4v^2}$
 (E) The ball will never land in the bucket.

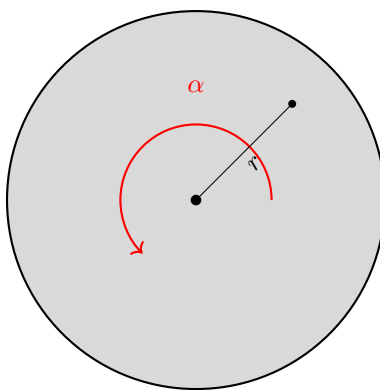
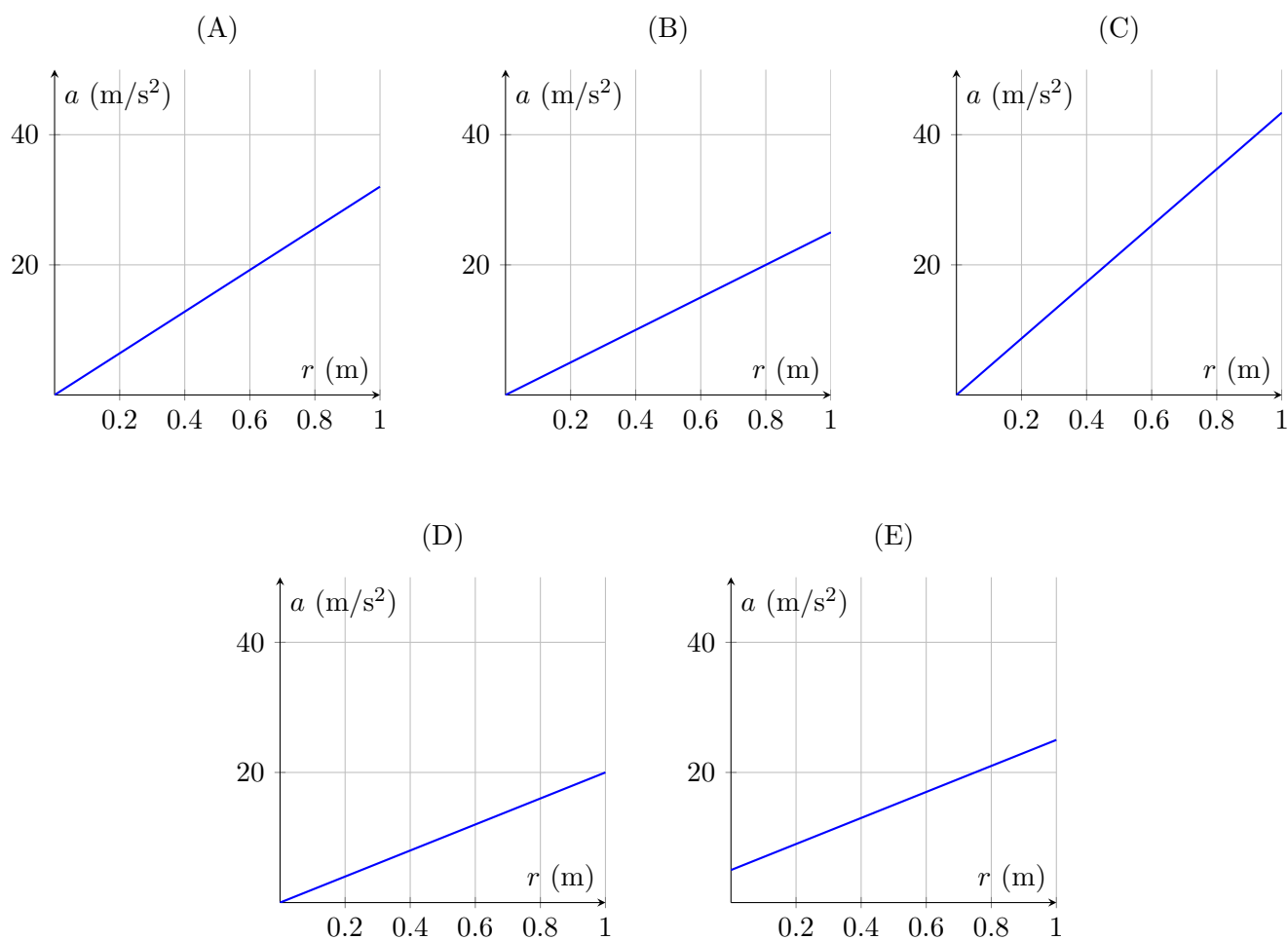
The time it takes the ball to reach the height of the bucket is $\frac{2v}{g}$. Let a be the acceleration of the train.

$$\frac{1}{2}a\left(\frac{2v}{g}\right)^2 = d$$

$$a = \frac{dg^2}{2v^2}$$

The answer is C.

6. A solid uniform disk rotates about a pivot through its center and perpendicular to its surface with constant angular acceleration $\alpha = 20 \text{ rad/s}^2$ starting from rest at $t = 0$. Which graph could represent the acceleration a at points on the disk a distance r away from its center as a function of r at $t = 0.25 \text{ s}$?

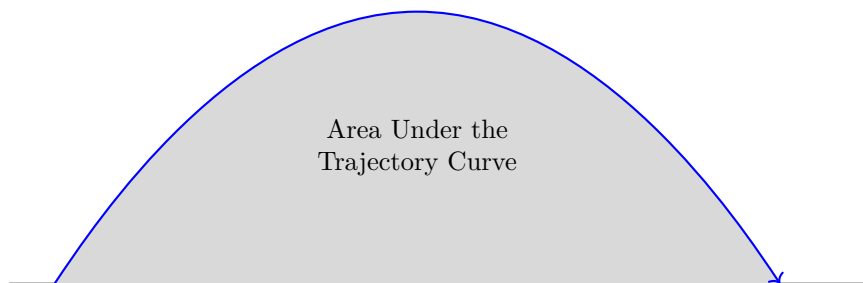
A solid disk rotating with angular acceleration α .

Let a_c and a_t be the centripetal and tangential accelerations, respectively. $a_c = \omega^2 r$ and $a_t = \alpha r$.

$$a(r) = \sqrt{a_c^2 + a_t^2} = r\sqrt{\omega^4 + \alpha^2} = r\sqrt{\alpha^4 t^4 + \alpha^2}$$

We select the linear graph going through the origin with slope $\sqrt{\alpha^4 t^4 + \alpha^2}$.

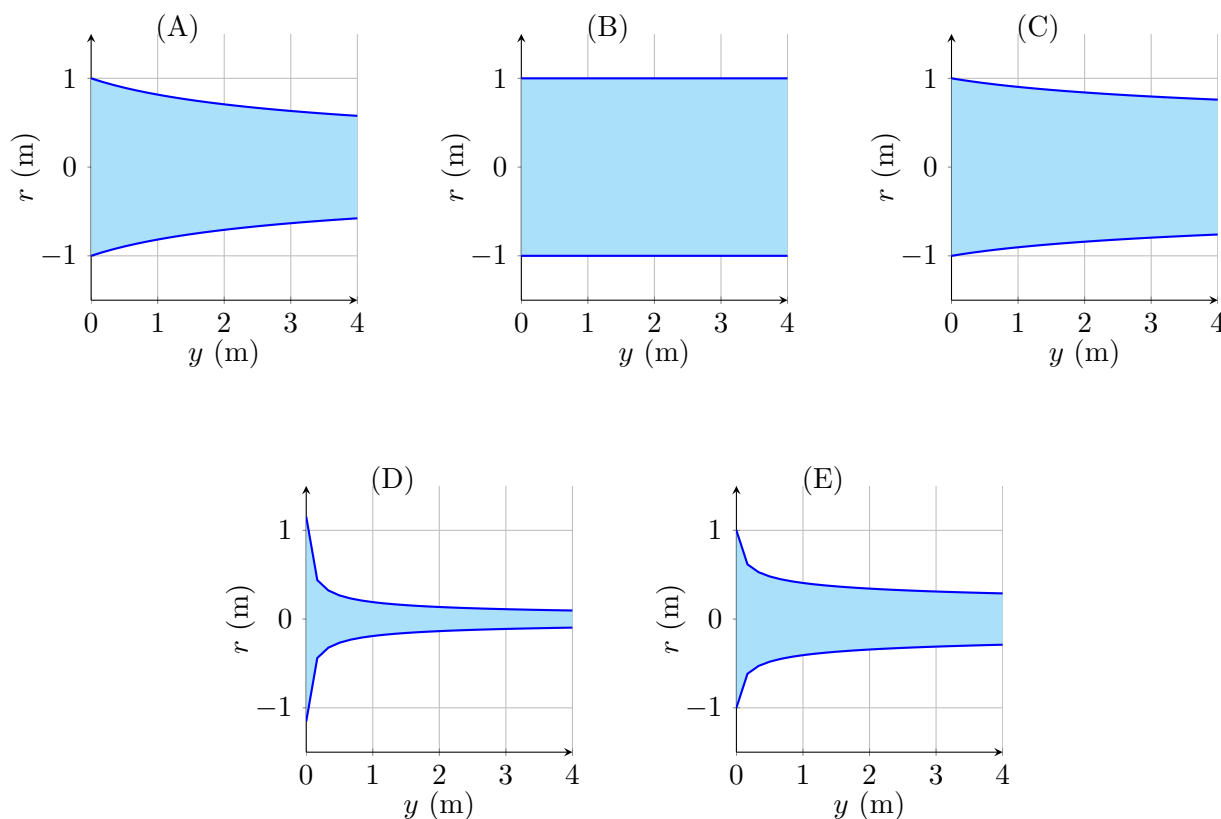
7. Projectile A is launched with an initial velocity v on a level surface at an angle θ above the horizontal on a uniform spherical planet which has mass M and radius R . Projectile B is launched with an initial velocity $4v$ on a level surface at an angle θ on a different uniform spherical planet which has mass $8M$ and radius $2R$. Let the areas under the trajectory curves of projectile A and projectile B respectively be A_A and A_B . Find A_B/A_A . Note that $v \ll \sqrt{\frac{GM}{R}}$.



- (A) 1 (B) 4 (C) 16 (D) 64 (E) 256

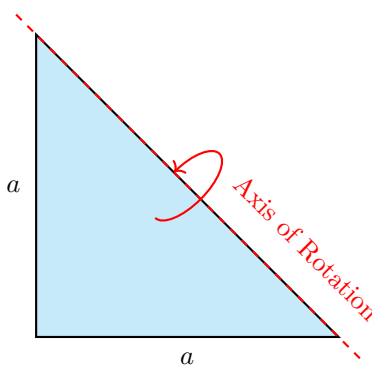
The area is proportional to $\frac{v^4}{g^2}$ where g is the local gravitational acceleration by dimensional analysis. g is equal to $\frac{Gm}{r^2}$ where m is the mass of the planet and r is the radius. The answer is thus 64.

8. Bob turns on a vertical faucet with a circular cross-section of radius 1 m. Which one of these diagrams best represents the shape of the stream as the water drops down? The speed of the water when it initially comes out of the faucet is 0.75 m/s. Note that the diagrams have been rotated for ease of viewing, i.e. you may consider gravity as pointing rightward in the graphs. You may neglect the component of the water's velocity along the r direction throughout its motion.



By continuity, Av is constant. We also note by energy conservation that $\frac{1}{2}\rho v_0^2 = \rho gy + \frac{1}{2}\rho v_y^2$. Combining this with the equation $\pi(1\text{ m})^2 v_0 = \pi r^2 v_y$ gives a graph that corresponds to answer E.

9. Calculate the moment of inertia of a planar uniform solid isosceles right triangle with legs of length a and mass m about the axis passing directly through the hypotenuse.

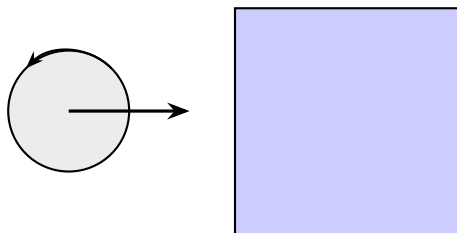


- (A) $\frac{1}{16}ma^2$ (B) $\frac{1}{12}ma^2$ (C) $\frac{1}{8}ma^2$ (D) $\frac{1}{6}ma^2$ (E) $\frac{1}{4}ma^2$

The moment of inertia of a uniform rod with mass m and length L about an axis through its center of mass perpendicular to the rod is $\frac{1}{12}mL^2$. "Stretch" the uniform rod into a uniform square of length L . The moment of inertia about that axis is still the same (the axis will now be through the perpendicular bisector of a side of the square). Then, using the perpendicular axis theorem and symmetry, the moment

of inertia through a diagonal of the square is equal to the moment of inertia through its perpendicular bisector ($\frac{1}{12}mL^2$). Finally, by symmetry, the moment of inertia of the given right triangle is equal to the moment of inertia through the diagonal of a square of equal mass and side length a , so the answer is $\frac{1}{12}ma^2$.

10. A solid uniform disk of mass m and radius $R = 0.8$ m lies flat on a frictionless surface. A square block of mass m and side length $l = 2$ m also lies flat on this surface at rest. As shown in the diagram, the disk is given an initial velocity $v_0 = 10$ m/s and angular velocity $\omega_0 = 30$ rad/s and collides head-on with the block so that it hits the midpoint of a side of the block. The coefficients of static and kinetic friction between the block and the disk are $\mu_s = \mu_k = 0.8$. If the disk's velocity vector at the instant after its first collision with the block is perpendicular to its initial velocity vector, find the speed of the block at that moment.



- (A) 10.00 m/s (B) 10.91 m/s (C) 11.66 m/s (D) 12.81 m/s (E) 15.62 m/s

Without loss of generality, let the disk's initial velocity be in the horizontal direction. The block receives a horizontal impulse mv_0 since the disk has no velocity in its initial direction after the collision. We must check if the disk begins rolling without slipping with respect to the block during the collision to analyze the vertical impulse. If the disk begins rolling without slipping, there will be no more frictional force between the disk and the block.

Let the total frictional impulse be J_f , the vertical velocity of the block be v_b , the angular velocity of the block be ω_B , and the vertical velocity of disk be v_d . For rolling without slipping to occur, J_f must be smaller than $mv\mu$. Note that the block begins rotating as well. To achieve rolling without slipping, there must be no relative velocity between the point of contact of the disk and the point of contact of the block:

$$\omega R - v_d = v_b + \omega_B l/2$$

Use angular impulse and linear impulse relationships to find

$$(\omega_0 - \frac{J_f R}{\frac{1}{2}mR^2})R - J_f/m = J_f/m + \frac{\frac{J_f l^2}{4}}{ml^2/6}$$

Thus,

$$J_f = \frac{2m\omega_0 R}{11}$$

We can see that this satisfies our condition for rolling without slipping, so J_f is the vertical impulse delivered to the block. The final answer is

$$\frac{\sqrt{J_f^2 + (mv_0)^2}}{m} = \sqrt{\left(\frac{2\omega_0 R}{11}\right)^2 + v_0^2} = 10.91 \text{ m/s}$$

11. A student creates a small droplet made out of an unknown liquid on a table, and slowly adds more liquid to the droplet until it breaks. If the density of the liquid is $(323 \pm 42.1) \text{ kg/m}^3$ and the surface tension between the liquid and air is $(3.75 \cdot 10^{-2} \pm 0.523 \cdot 10^{-2}) \text{ N/m}$, find the percent error in calculating the volume of the droplet right before it breaks. Assume that the values of all other relevant quantities are known to negligible uncertainty, and that the surface tensions between the table and air or liquid are 0.

(A) 19.1% (B) 26.0% (C) 28.6% (D) 51.9% (E) 77.9%

By dimensional analysis, the volume is proportional to $(\frac{\gamma}{\rho g})^{1.5}$. Thus, the percent uncertainty is $1.5\sqrt{(\frac{42.1}{323})^2 + (\frac{0.523}{3.75})^2} = 28.6\%$.

12. Consider a flat two-dimensional Cartesian plane. A point mass is placed at rest at the origin. At $t = 0$, the particle begins accelerating. Its acceleration vector has constant magnitude $a = 10\pi \text{ m/s}^2$, rotates counterclockwise with constant angular velocity $\omega = 5\pi \text{ rad/s}$, and initially points in the x direction. Find the position of the particle at $t = 100 \text{ s}$.

(A) (0, 100 m) (B) (100 m, 0) (C) (0, 200 m) (D) (200 m, 0) (E) (100 m, 100 m)

Work in the frame moving at $v = a/\omega$ in the positive y -direction. In this frame, the point mass is in uniform circular motion, returning to the origin after each period. Each period is $\frac{2\pi}{\omega} = \frac{2}{5} \text{ s}$. Since $t = 100 \text{ s}$ is an integer multiple of the period, the point mass is at the origin in the moving reference frame at that time. Thus, the final position of the particle is $(0, \frac{at}{\omega}) = (0, 200 \text{ m})$ in the lab frame.

13. A regular uniform n -gon (a regular polygon with n sides) of side length a and mass m is held at rest on an incline angled at 25° above the horizontal. If the coefficient of static friction between the incline and the n -gon is $\mu = 0.8$, find the largest n such that the n -gon will remain in equilibrium after being released from rest.

(A) 4 (B) 5 (C) 6 (D) 7 (E) 8

The n -gon will not slide down the incline since $\tan(25^\circ) < \mu$. The distance between the center of mass of the n -gon and one of its sides is $\frac{a}{2 \tan(\frac{180^\circ}{n})}$. At the critical point, all of the normal force will be concentrated on one vertex of the n -gon, so we take torques about that point.

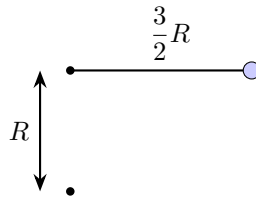
$$mg \sin(\theta) \cdot \frac{a}{2 \tan(\frac{180^\circ}{n})} = mg \cos(\theta) a/2$$

$$\tan \theta = \tan\left(\frac{180^\circ}{n}\right)$$

Solving, we get $n = 7.2$, so $n = 7$ is the largest n that works.

14. Consider two infinitesimally small pivots on a frictionless wall fixed a vertical distance R apart. A massless string of length $\frac{3}{2}R$ is attached to the top pivot, and a small ball is attached to the end of the string. The ball is held perfectly horizontally and released from rest. Starting from the string's collision with the

bottom pivot, find the angle that the ball swings through relative to the bottom pivot before the string loses tension.



- (A) 108°
- (B) 139°
- (C) 153°
- (D) 167°
- (E) The string does not lose tension.

By energy conservation, the velocity of the ball when the string collides with the bottom pivot is $\sqrt{2g \cdot \frac{3}{2}R} = \sqrt{3gR}$. By angular momentum conservation about the bottom pivot, the speed does not change during the collision, so energy is conserved. Let θ be the angle from the vertical at which the string loses tension. At the critical point,

$$\frac{mv^2}{R/2} = mg \cos \theta$$

$$\frac{1}{2}m \cdot 3gR = \frac{1}{2}mv^2 + \frac{mgR}{2}(1 + \cos \theta)$$

Combining the equations,

$$\frac{3}{2}mgR = \frac{mgR \cos \theta}{4} + \frac{mgR}{2}(1 + \cos \theta)$$

Solving, we see that

$$\cos \theta = \frac{4}{3}$$

No θ exists, so the string does not lose tension.

15. Consider an incline of mass m angled at θ above the horizontal with a height h placed at rest on top of a frictionless floor. A small block of mass $m/10$ is held at rest at the very top of the incline and released. The block slides down the incline and comes to a complete stop just as it reaches the bottom of the incline due to friction between the incline and the block (The block does not touch the floor). Find the magnitude of the displacement of the block relative to the ground.

- (A) $\frac{h}{\sin \theta}$
- (B) $h\sqrt{1 + \left(\frac{10 \cot \theta}{9}\right)^2}$
- (C) $h\sqrt{1 + \left(\frac{9 \cot \theta}{10}\right)^2}$

- (D) $h\sqrt{1 + (\frac{11\cot\theta}{10})^2}$
 (E) $h\sqrt{1 + (\frac{10\cot\theta}{11})^2}$

Obviously, the y-displacement of the block is h . We find the x-displacement by conserving the center of mass of the block-incline system. Let x_1 be the horizontal displacement of the block and let x_2 be the horizontal displacement of the incline.

$$x_1 + x_2 = \frac{h}{\tan\theta}$$

$$\frac{mx_1}{10} = mx_2$$

$$\frac{11x_1}{10} = \frac{h}{\tan\theta}$$

Our final answer is

$$\sqrt{h^2 + (\frac{10h}{11\tan\theta})^2} = h\sqrt{1 + (\frac{10\cot\theta}{11})^2}$$

16. Consider the following statements if the gravitational force between two masses m_1 and m_2 at a distance r apart was equal to $\frac{Gm_1m_2}{r}$, still being an attractive force as usual, with the units of G adjusted accordingly.
- I: The gravitational field at any point inside a uniform spherical shell of nonzero mass is zero.
- II: The gravitational field at any point in the plane of a uniform ring of nonzero mass, with the point completely surrounded by the ring, is zero.

Consider the following statements if the gravitational force between two masses m_1 and m_2 at a distance r apart was equal to $\frac{Gm_1m_2}{r^3}$, still being an attractive force as usual, with the units of G adjusted accordingly.

- III: The gravitational field at any point inside a uniform spherical shell of nonzero mass is zero.
- IV: The gravitational field at any point inside a uniform sphere of nonzero mass is zero.

Given their corresponding scenarios, which statements are true?

- (A) II only.
 (B) IV only.
 (C) I and III only.
 (D) II and IV only.
 (E) None of the statements are true.

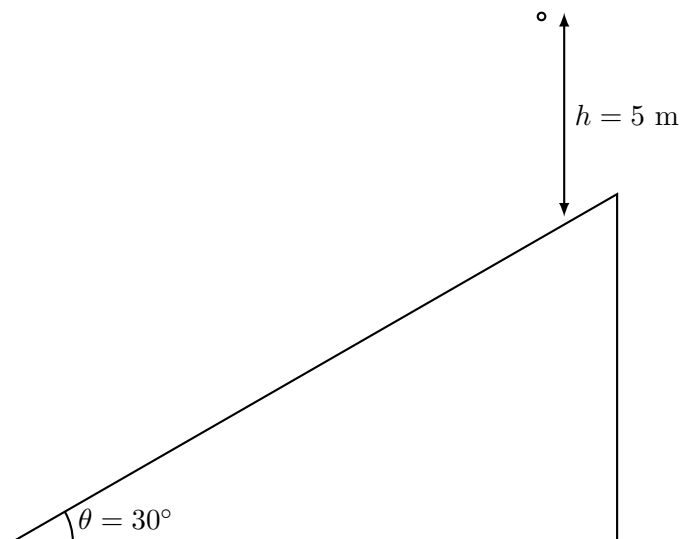
I and III are wrong following the standard derivation for the shell theorem:

Consider any point in the shell and draw two lines through the point, making a small angle θ . The lines will intersect the shell. Consider the cone bounded by the lines, and the small area element at the end of the cone on the shell. The area element will be proportional to r^2 , and multiplying by the

density of the shell gives its mass proportional to r^2 . This only cancels out with the gravitational force for gravity proportional to r^{-2} .

II is true following similar logic. Now, we show that IV is false. To see why, use the same model. In this case, we will have solid cones as opposed to area elements. Imagine cutting up the cones such that they are composed of many cylinders of very small thickness Δr . Now, following the same argument as before, the gravitational force due to each of these thin cylinders is proportional to $\frac{r^2 \Delta r}{r^3} = \frac{\Delta r}{r}$. It is evident that the the sum of the gravitational forces due to the cylinders on either side will not cancel out in this case unless the cones are of equal length, which is not the case unless the point chosen is at the center of the sphere. It is also evident that the net force due to all of these small forces will not cancel out by considering some cases (ex: a point close to the boundary of the sphere). Thus, our final answer is A.

17. A small ball is released at a height $h = 5$ m above a long fixed frictionless incline angled at 30° above the horizontal. Let point A be the point of initial contact between the small and the incline. The ball has a coefficient of restitution $\alpha = 0.90$ with the incline. Let point B be the point at which the ball stops bouncing on the incline and begins sliding down. Find the distance between point A and point B. The incline is long enough such that point A and point B are both on the incline. The diagram is not to scale.



- (A) 600 m (B) 700 m (C) 800 m (D) 900 m (E) 1000 m

Let v_n denote the magnitude of the component of the velocity of the ball perpendicular to the slope immediately after the n th collision with the slope. We can see that

$$v_1 = \alpha \sqrt{2gh} \cos \theta$$

$$v_2 = \alpha^2 \sqrt{2gh} \cos \theta$$

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$$v_n = \alpha^n \sqrt{2gh} \cos \theta$$

The time of flight after the n th collision is

$$t_n = \frac{2v_n}{g \cos \theta} = \frac{2\alpha^n \sqrt{2gh}}{g}$$

Thus, we can see that the total time of flight for the ball after its initial collision with the incline is

$$t_{\text{total}} = \frac{\frac{2\alpha \sqrt{2gh}}{g}}{1 - \alpha}$$

The distance between point A and point B is

$$\frac{1}{2}g \sin(\theta)t_{\text{total}}^2 + \sqrt{2gh} \sin(\theta)t_{\text{total}}$$

Plugging in the numbers, we get our final result of 900 m.

A satellite is in uniform circular motion around a planet with an orbital radius R and speed v_i . Suddenly, it applies its rocket boosters for an instantaneously short period of time, increasing its speed only in the direction of its motion. Its speed immediately after the boost is v_f . The maximum distance between the rocket and the planet in its new orbital trajectory is $5R$. Find v_f/v_i .

- (A) $\sqrt{2}$ (B) $\frac{2}{\sqrt{3}}$ (C) $\frac{4}{3}$ (D) $\sqrt{\frac{5}{3}}$ (E) $\frac{5}{3}$

18. We use the expression for the energy of an object in an elliptical orbit of semimajor axis a , which is $-\frac{GMm}{2a}$. We have

$$-\frac{GMm}{2R} = -\frac{GMm}{R} + \frac{1}{2}mv_i^2$$

$$-\frac{GMm}{6R} = -\frac{GMm}{R} + \frac{1}{2}mv_f^2$$

$$\frac{GMm}{3R} = \frac{1}{2}m(v_f^2 - v_i^2)$$

$$\sqrt{\frac{2GM}{3R}} + v_i^2 = v_f^2$$

$$\frac{v_f}{v_i} = \frac{\sqrt{\frac{2GM}{3R}} + v_i^2}{v_i}$$

We also know that

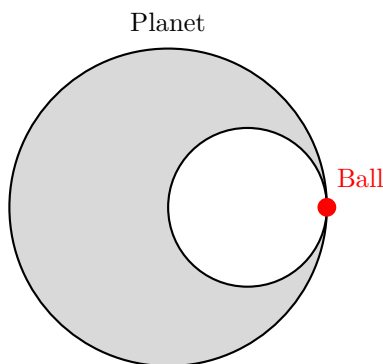
$$\frac{mv_i^2}{R} = \frac{GMm}{R^2}$$

$$\frac{GM}{R} = v_i^2$$

Thus, our answer is

$$\sqrt{\frac{5}{3}}$$

19. Consider a uniform spherical planet of mass m and radius r . A spherical cavity of radius $r/2$ is cut out of the planet, which intersects the surface and the center of the planet. A small ball is placed at the intersection of the planet's surface and the cavity and released from rest. Find the time t it takes for the ball to collide with the center of the planet.



- (A) $\sqrt{\frac{r^3}{Gm}}$ (B) $\sqrt{\frac{2r^3}{Gm}}$ (C) $2\sqrt{\frac{r^3}{Gm}}$ (D) $2\sqrt{\frac{2r^3}{Gm}}$ (E) $4\sqrt{\frac{r^3}{Gm}}$

By vector addition, the acceleration is uniform in the cavity (to show this, use the shell theorem for a point inside a sphere, note that the gravitational field is proportional to the distance from the sphere's center, and sum the gravitational field of the planet with the cavity filled in with the negative mass field due to the cavity). Thus, we may calculate the acceleration at a convenient point in the cavity, and then use that for all points. At the center of the planet,

$$\frac{G \cdot \frac{m}{8}}{r^2/4} = a$$

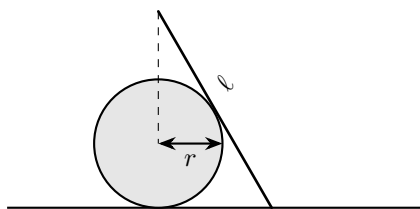
$$a = \frac{Gm}{2r^2}$$

Finding the time t :

$$\frac{1}{2}at^2 = r$$

$$t = \sqrt{\frac{2r}{a}} = 2\sqrt{\frac{r^3}{Gm}}$$

20. A cylinder is fixed to the ground. A uniform stick rests on the cylinder so that the top of the stick is positioned above the cylinder's center, as shown in the diagram. The stick makes an angle $\theta = 50^\circ$ with the horizontal. Given that the stick and cylinder have negligible friction with each other, find the coefficient of static friction μ between the stick and the ground necessary to keep the system in equilibrium. Note that the stick's corners are slightly rounded.



- (A) 0.386 (B) 0.499 (C) 0.538 (D) 0.564 (E) 0.839

Since the stick's corners are slightly rounded, the normal force between the ground and the stick points perpendicular to the ground. If we do some geometry, we find that the distance between the top of the rod and its contact point with the cylinder is $R \tan \theta$. Let N_1 be the normal force between the cylinder and the stick and let N_2 be the normal force between the stick and the ground. Balancing torques about the bottom point,

$$N_1(l - R \tan \theta) = \frac{mgl}{2} \cos \theta$$

Balancing forces in the x and y directions:

$$mg = N_2 + N_1 \cos \theta$$

$$N_1 \sin \theta = N_2 \mu$$

Solving,

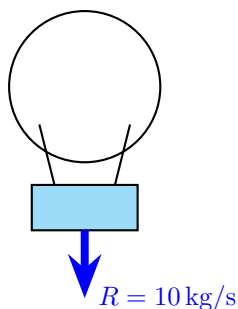
$$N_1 = \frac{mgl \cos \theta}{2(l - R \tan \theta)}$$

$$N_2 = mg - N_1 \cos \theta = mg - \frac{mgl \cos^2 \theta}{2(l - R \tan \theta)}$$

$$\mu = \frac{N_1 \sin \theta}{N_2} = \frac{\frac{mgl \cos \theta}{2(l - R \tan \theta)} \sin \theta}{mg - \frac{mgl \cos^2 \theta}{2(l - R \tan \theta)}} = \frac{l \sin \theta \cos \theta}{2(l - R \tan \theta) - l \cos^2 \theta}$$

From geometry, we have that $l = R \tan \theta + R \cot(\theta/2)$. Plugging in this result into our above equation and calculating gives $\mu = 0.564$.

21. A hot air balloon is initially floating in the air at rest and carries a tank of water of mass $m = 10^4$ kg in its basket. At $t = 0$, the balloon begins slowly leaking water at a constant rate of $R = 10$ kg/s. Assume that the hot air balloon's volume does not change, and neglect all energy losses due to friction or drag. The hot air balloon's mass is negligible. Which of the following answer choices is closest to the speed of the balloon at $t = 5$ s? Hint: You may use the approximation $(1 + x)^n \approx 1 + xn$ for $|xn| \ll 1$.



- (A) 0.13 m/s (B) 0.25 m/s (C) 0.38 m/s (D) 0.50 m/s (E) 0.68 m/s

The buoyant force on the balloon is mg . The key to solving this problem is to realize that the acceleration of the balloon is changing with time, and to make an analogy with the standard kinematic equation $x = x_0 + v_0t + \frac{1}{2}at^2$ to find that $v = v_0 + a_0t + \frac{1}{2}jt^2$, for a constant j , where j is the jerk (the rate of change of the acceleration). The acceleration of the balloon can be found using $F = ma$:

$$mg - (m - Rt)g = (m - Rt)a$$

$$a = \frac{Rtg}{m - Rt} \approx \frac{Rtg}{m} \left(1 + \frac{Rt}{m}\right) \approx \frac{Rtg}{m}$$

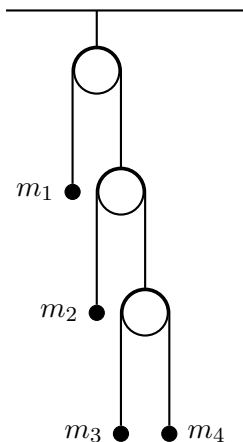
Note that we have dropped the second order term in the above expression. To see why we can do this, $Rt/m = 10^{-2}$, so $(Rt/m)^2 = 10^{-4}$. The second order term is one percent of the first order term and thus has a negligible effect on the acceleration at the precision that we are working.

Thus, we can see that the jerk is $\frac{Rg}{m}$. Plugging this into our equation, we find that

$$v = \frac{Rg}{2m}t^2 = 0.125 \text{ m/s}$$

As a sidenote, an exact solution by integrating the acceleration gives $v = 0.1254 \text{ m/s}$.

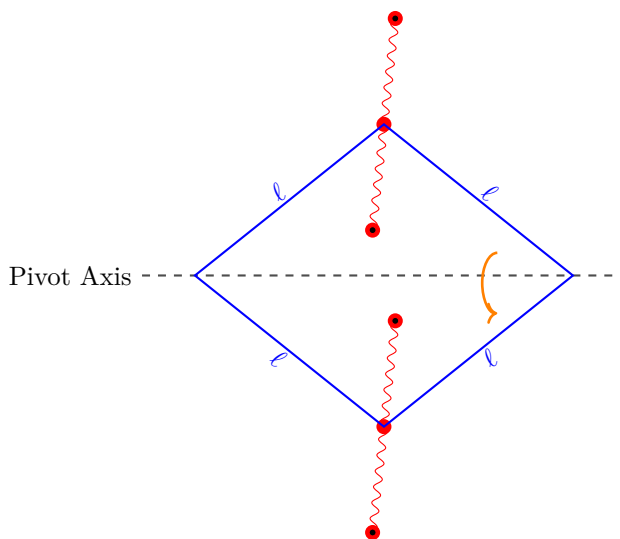
22. In the pulley system below, find the acceleration of m_1 in terms of m_2 , m_3 , m_4 , and g . All pulleys and strings are massless and frictionless. Hint: The algebra required to find the correct answer is very minimal.



- (A) $g \left(\frac{\frac{8}{m_1} - \frac{1}{m_2} - \frac{1}{m_3} - \frac{1}{m_4}}{\frac{8}{m_1} + \frac{1}{m_2} + \frac{1}{m_3} + \frac{1}{m_4}} \right)$
- (B) $g \left(\frac{\frac{8}{m_1} - \frac{4}{m_2} - \frac{1}{m_3} - \frac{1}{m_4}}{\frac{8}{m_1} + \frac{4}{m_2} + \frac{1}{m_3} + \frac{1}{m_4}} \right)$
- (C) $g \left(\frac{\frac{16}{m_1} - \frac{2}{m_2} - \frac{1}{m_3} - \frac{1}{m_4}}{\frac{16}{m_1} + \frac{2}{m_2} + \frac{1}{m_3} + \frac{1}{m_4}} \right)$
- (D) $g \left(\frac{\frac{16}{m_1} - \frac{4}{m_2} - \frac{1}{m_3} - \frac{1}{m_4}}{\frac{16}{m_1} + \frac{4}{m_2} + \frac{1}{m_3} + \frac{1}{m_4}} \right)$
- (E) $g \left(\frac{\frac{8}{m_1} - \frac{2}{m_2} - \frac{1}{m_3} - \frac{1}{m_4}}{\frac{8}{m_1} + \frac{2}{m_2} + \frac{1}{m_3} + \frac{1}{m_4}} \right)$

This problem is intended to be solved with limiting cases. We let $m_3 = m_4 = m$ and $m_2 = 2m$. Now, this system is equivalent to having a mass $4m$ on one side of a frictionless pulley and a mass m_1 on the other side. This should yield an acceleration of $g \frac{4m - m_1}{m_1 + 4m}$, which is indeed what we find for answer choice D, and none of the other answer choices.

23. A square is constructed by attaching four uniform rigid rods, each of mass m and length l together at their endpoints. The square is then pivoted about an axis passing through two opposite corners. Four springs, each of spring constant k are attached to the other two corners such that each of the springs are perpendicular to the plane of the square and each corner has two springs attached to it, with one on each side. The springs are all initially at their relaxed length and the square is at rest. The square is then rotated by a small angle about the aforementioned axis. Find the period of small oscillations and neglect gravity.



- (A) $2\pi\sqrt{\frac{m}{3k}}$ (B) $2\pi\sqrt{\frac{5m}{6k}}$ (C) $2\pi\sqrt{\frac{2m}{3k}}$ (D) $2\pi\sqrt{\frac{5m}{3k}}$ (E) $2\pi\sqrt{\frac{m}{6k}}$

The springs are at a distance $l/\sqrt{2}$ from the axis. Thus, the restoring torque is $4k\theta(l/\sqrt{2})^2 = 2kl^2\theta$. The moment of inertia about the pivot can be found by finding the moment of inertia about the axis passing through the center of mass of the square perpendicular to its plane, and then using the perpendicular axis theorem. Each rod's center of mass is $\frac{l}{2}$ away from the square's center of mass. Then, using the parallel axis theorem,

$$I_{\text{cm, perp}} = 4\left(\frac{1}{12}ml^2 + m\left(\frac{l}{2}\right)^2\right) = \frac{4}{3}ml^2$$

By the perpendicular axis theorem, the moment of inertia about the pivot that we care about is half of this. Finally,

$$\frac{2}{3}ml^2\alpha = -2kl^2\theta$$

$$\omega = \sqrt{\frac{3k}{m}}$$

$$T = 2\pi\sqrt{\frac{m}{3k}}$$

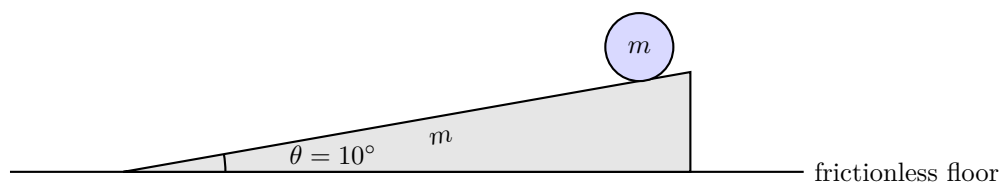
24. A block is placed at rest on a frictionless floor and attached to a massless spring of spring constant k and nonzero relaxed length l . The mass is given a push so that it begins oscillating in simple harmonic motion with amplitude $A < l$. After a while, wind begins to blow in a constant direction with a constant speed directly along the block's path. The wind imparts a force $b(\vec{v} - \vec{v}_b)$ to the block, where $\vec{v}$ is the (constant) velocity vector of the wind and $\vec{v}_b$ is the (changing) velocity vector of the block. Assume that $b|\vec{v}| \ll kA$. After a long time, how does the speed of the wind impact the amplitude of the block's oscillations?



- (A) The amplitude does not change for any value of $|\vec{v}|$.
- (B) The amplitude decreases for all nonzero $|\vec{v}|$.
- (C) The amplitude increases for all nonzero $|\vec{v}|$.
- (D) The amplitude decreases for all nonzero $|\vec{v}| > x$ and increases for all nonzero $|\vec{v}| < x$, where x is a nonzero finite constant with units of m/s.
- (E) The amplitude increases for all nonzero $|\vec{v}| > x$ and decreases for all nonzero $|\vec{v}| < x$, where x is a nonzero finite constant with units of m/s.

The amplitude decreases for all nonzero $|\vec{v}|$. To see this, the wind provides a positive power to the block only when the velocity of the block points in the same direction as the wind with a lower magnitude than the wind's velocity. This does not occur at all for half of the period of the block's motion, and occurs only for two separate intervals for the other half. Thus, by symmetry, the net power delivered to the block is negative, decreasing its amplitude.

25. A uniform cylinder of mass m is held at rest at the top of an incline angled at $\theta = 10^\circ$, also of mass m , which lies at rest on top of a frictionless floor. The cylinder is released from rest. Find the minimum coefficient of static friction μ between the cylinder and the incline such that the cylinder immediately begins rolling without slipping down the incline when it is released.



- (A) 0.0441
 (B) 0.0882
 (C) 0.176
 (D) 0.353
 (E) Achieving rolling without slipping is impossible, regardless of how large μ is.

Work in the accelerating frame of the incline. Let N be the normal force, a be the acceleration of the incline, a_c be the acceleration of the cylinder, and θ be the angle. Balance horizontal forces on the incline:

$$N \sin \theta = ma + N\mu \cos \theta$$

Use Newton's second law for the horizontal and vertical acceleration of the cylinder relative to the inclined surface:

$$ma \sin \theta + N = mg \cos \theta$$

$$ma \cos \theta + mg \sin \theta - N\mu = ma_c$$

Relate the torque to the acceleration using the rolling without slipping condition:

$$N\mu R = \frac{1}{2}mR^2\alpha$$

$$a_c = \frac{2N\mu}{m}$$

Combine the fourth equation with the third equation:

$$ma \cos \theta + mg \sin \theta - N\mu = 2N\mu$$

$$3N\mu = ma \cos \theta + mg \sin \theta$$

Eliminate ma using the first equation:

$$3N\mu = N \sin \theta \cos \theta - N\mu \cos^2 \theta + mg \sin \theta$$

$$N \sin^2 \theta - N\mu \sin \theta \cos \theta + N = mg \cos \theta$$

After eliminating N from the above two equations, we obtain

$$\mu = \frac{\sin \theta + \sin^3 \theta + \sin \theta \cos^2 \theta}{3 \cos \theta + \cos^3 \theta + \sin^2 \theta \cos \theta} = \frac{1}{2} \tan \theta = 0.0882$$